Perceptual Robust Hashing for Video Copy Detection with Unsupervised Learning

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ABSTRACT

In this paper, we propose an end-to-end perceptual robust hashing scheme for video copy detection based on unsupervised learning. Firstly, the spatio-temporal information in videos is effectively fused and condensed into high-dimensional features through a 3D self-attention, multi-scale feature fusion model based on 3D-CNN, in which the Inception block and the 3D self-attention mechanism are integrated. Then, we calculate the correlation distances between the extracted features to differentiate perceptual contents. Based on the similarity relationship, we can dynamically generate the pseudo-labels and exploit them to further guide the model training for video hash generation. In addition, we design the dual constraints to make the hash code obtain satisfactory robustness and discrimination. Extensive experiments demonstrate that the proposed scheme achieves superior performance of copy detection compared with existing schemes and performs well even in the case of untrained manipulations.

CCS CONCEPTS

• **Security and privacy → Database and storage security**.

KEYWORDS

perceptual video hashing; 3D-CNN; unsupervised learning; copy detection

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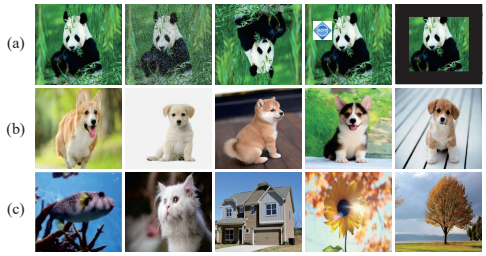


Figure 1: Example illustration of sample similarity. (a) Perceptual similar samples. (b) Semantic similar samples. (c) Irrelevant samples.

1 INTRODUCTION

With the development of multimedia technology, a large number of videos are transmitted on social media platforms. In practice, the same video may be uploaded by different users and pirates, and the video may also undergo common manipulations, such as compression and watermark embedding. The file sizes and formats of the videos may be different, but they have perceptual identical contents. Thus, thousands of copies of a video may exist, resulting in copyright infringement and storage wasting. The technique of perceptual hashing [1] can map multimedia data into a short sequence, which is equivalent to the fingerprint of multimedia data, known as hash code, and can be used in the fields of copy detection [2], content authentication [3], tampering detection [4], etc.

Inspired by several excellent image hashing schemes [5–7], some researchers extended the image hashing directly to the video hashing. Yang *et al.* [2] divided the video into frames, calculated the speed up robust feature (SURF) of each frame, and used the Hilbert curve to count the number of feature points to obtain the hash code. A robust video hashing scheme for content authentication was proposed by Lee and Yoo [8], and this scheme constructed a hash sequence by segmenting each video frame and calculating the centroid of gradient orientations (CGO), which was sensitive to frame rotation and cropping.

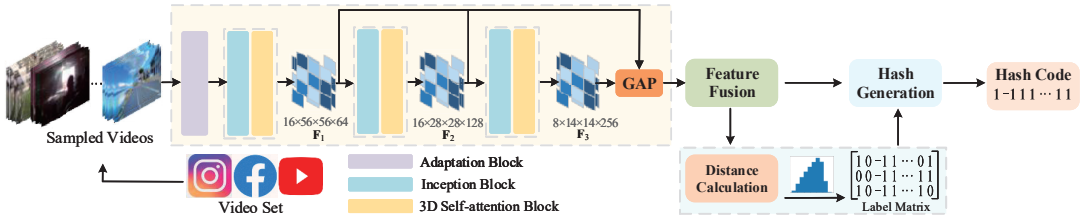


Figure 2: Overall framework of our unsupervised perceptual video hashing scheme.

However, unlike image hashing, video hashing encounters great challenges because videos contain spatio-temporal information [9, 10], such as human actions, background transformation and some high-level perceptual features, which make it difficult to capture the key information of videos. To this end, Setyawan *et al.* [11] proposed a spatio-temporal domain hashing scheme based on edge orientation histogram (EOH) and discrete cosine transform (DCT). The scheme first computed the EOH of each frame in the video, and then performed a temporal DCT on the EOH of the same direction to obtain the spatio-temporal domain feature. Saikia [12] performed 3D-DWT on the video and extracted the coefficients from the low-pass sub-bands, and then conducted DCT on these coefficients to obtain the hash code. Sandeep *et al.* [13] extended the two-dimensional radial projection technique (RPT) for image hashing to three dimensions by calculating the variance of the luminance values of the projected pixels for each sub-tensor, and then applied DCT to generate a video hash sequence.

The above schemes divided feature extraction and hash mapping into two parts, and both were designed artificially and empirically, which may not achieve the global optimal result. Therefore, the defects of existing schemes has triggered researchers to investigate new hash generation methods. Numerous studies [14, 15] have shown that deep learning techniques can effectively extract the features of videos as long as a proper model and loss function are designed. Therefore, we can build an end-to-end scheme based on deep learning that integrates feature extraction and hash mapping to generate perceptual spatio-temporal feature of the video. But, current perceptual hashing schemes often require manual annotation of video samples and their similarity relationships, and these methods may not have enough generalization capability to deal with massive video and emerging video manipulations [16]. To address these issues, in this work, we propose a perceptual robust video hashing scheme based on unsupervised learning.

Main contributions of this work can be summarized as follows:

- An end-to-end unsupervised perceptual video hashing scheme is proposed that can be used for content authentication and copy detection.
- In order to represent the video effectively, we design a 3D self-attention multi-scale feature fusion model, which can extract compact perceptual feature and highlight the important parts. In addition, the model training can be better guided and constrained due to the employment of pseudo-labels.
- Our scheme can not only achieve satisfactory robustness and discrimination, but also adapts to more video content-preserving manipulations without labeling and training.

2 MOTIVATION

2.1 Challenge

Unlike the content-based image/video retrieval [17, 18], which mainly focuses on semantic level retrieval by identifying the object class in the image or video corresponding to the query one, the perceptual robust hashing concerns more about the perceptual features [19] in the image or video to identify the same image/video under the condition of different content-preserving manipulations, see Fig. 1(a), rather than the target subject with the same semantic class, see Fig. 1(b). Therefore, the task of perceptual robust hashing is not easy because of the poor interpretability of neural networks and the difficulty of how to learn perceptual content features robust against content-preserving manipulations.

Detailedly, the task of perceptual robust video hashing is to identify perceptual identical videos and differentiate perceptual distinct videos. For a given original video, its perceptual identical versions can be defined as various versions after undergoing content-preserving manipulations, such as enhancement, scaling, logo insertion and re-sampling, which can be also called as copied versions; while its perceptual distinct versions can be considered as the completely irrelevant videos, see Fig. 1(c), even including the videos with the same semantic class as shown in Fig. 1(b).

2.2 Problem Definition

Based on the above definition, we can summarize the problem as follows: given an original video v, a video v^+ whose perceptual content is identical with v, and a video v^- whose perceptual content is different from v. The objective of a robust hashing scheme is to construct a transform $f(\cdot)$: $v \rightarrow \mathbf{c}$, where $\mathbf{c} \in \{-1, 1\}^l$ from video space to a l-bits hash code. An ideal robust hashing scheme should satisfy the following two properties: a) *Perceptual Robustness*: the hash codes transformed from videos with the same perceptual content should be the same or similar; b) *Discrimination*: the hash codes of videos with different perceptual contents should produce significant variations. The properties can be summarized as:

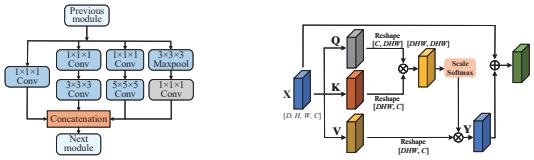
$$\begin{cases} P(D(f(v), f(v^+)) < T) \geq 1 - \varepsilon \\ P(D(f(v), f(v^-)) \geq T) \geq 1 - \varepsilon \end{cases} \tag{1}$$

where $P(\cdot)$ and $D(\cdot, \cdot)$ denote the probability of event occurrence and the hash distance, respectively, T is the judgment threshold, and ε is the extremely small positive number.

3 PROPOSED SCHEME

3.1 Model Description

The flowchart of our unsupervised hashing scheme is given in Fig. 2. Firstly, we extract and fuse spatio-temporal features of videos



(a) Inception block (b) 3D self-attention block

Figure 3: Illustration of Inception and 3D self-attention blocks.

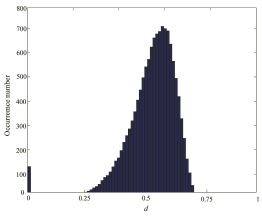


Figure 4: The histogram of correlation distances d.

by an efficient 3D-CNN model. Then, the extracted video-level information are statistically analyzed to generate a pseudo-label matrix. Finally, we output the values of the last fully connected layer (FC) and binarize them to form the hash code.

Specifically, the adaptation block is used to pre-process each input video into 32-frame clips, with each frame of size 224×224×3, and the superficial features are extracted by a 3×3×3 convolution. These features are fed into the Inception blocks for further analysis, which can extract different granularities of features. Inspired by the 2D InceptionNet [20], we design an improved Inception block, which can not only extract the spatial information of the frame, but also capture the long-term dependencies of frame sequence, as shown in Fig. 3(a). The block consists of four main branches, including the first branch with a 1×1×1 convolution, the second branch with a 1×1×1 convolution and a 3×3×3 convolution, the third branch with a 1×1×1 convolution and a 5×5×5 convolution, and the fourth branch with a 1×1×1 convolution followed by a maxpooling. We keep the output feature maps of different branches as the same size and concatenate them together. To accelerate the convergence of the model, each convolution is followed by batch normalization (BN) and activation function (Leaky ReLU). The operation of the Inception block can be formulated as:

$$y = Concat[\zeta^1(x); \zeta^2(x); \zeta^3(x); \zeta^4(x)] \tag{2}$$

where ζ^i, $i = 1, 2, 3, 4$, denote the different processing of the four branches, x and y are the input and output of the Inception block, respectively, and $Concat(\cdot)$ represents the concatenation operation. The combination of branches can analyze the features from different receptive fields and enhance the adaptability of the model to fuse multi-scale features.

In order to generate more representative video feature, we design the 3D self-attention mechanism to highlight the important parts, as shown in Fig. 3(b). Given the input feature $\mathbf{X}\in\mathbb{R}^{F\times H\times W\times C}$, where F denotes the number of feature maps, H, W and C denote the height, width and channel of the feature map, respectively. Then, after passing through three 1×1×1 convolutions, the input feature $\mathbf{X}$ is transformed into three different matrices, i.e., $\mathbf{Q}\in\mathbb{R}^{C\times r}$, $\mathbf{K}\in\mathbb{R}^{r\times C}$ and $\mathbf{V}\in\mathbb{R}^{r\times C}$, respectively, where $r = FHW$. Next, we analyze the inherent relationship in $\mathbf{X}$ to generate the attention score by a softmax function, and map this score to the matrix $\mathbf{V}$ as follows:

$$\mathbf{Y} = Softmax\left(\frac{\mathbf{Q}\cdot\mathbf{K}}{\sqrt{r}}\right)\mathbf{V} \tag{3}$$

where $\mathbf{Y}\in\mathbb{R}^{r\times C}$ is the emphasized feature map. Then, we reshape $\mathbf{Y}$ and perform element-wise addition with the input feature map $\mathbf{X}$ through a residual connection structure. Due to the 3D self-attention mechanism, the information within the feature map can be correlated to each other and highlight more important parts. We place this mechanism behind the Inception blocks to obtain more robust perceptual feature.

Because of the stacking of Inception blocks, some superficial perceptual information may be lost in the deeper block. To prevent the loss of key information, we integrate features across blocks. Detailedly, for the three output feature maps with different scales in Fig. 2 (i.e., $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$), we perform the global average pooling (GAP) operation on them in the channel dimension separately, and concatenate them together to form a vector size of 1×1×1×448 (i.e., $64 + 128 + 256 = 448$). Then, the vector is dimensionally reduced by a 1×1×1 convolution kernel to obtain the perceptual feature of the entire video, denoted as $\mathbf{Z}$.

After that, we conduct the analysis on statistical distribution of video-level features $\mathbf{Z}$ in the batch, and generate pseudo-labels based on the perceptual similarity to guide the training of the model (Details are described in Section 3.2). In the module of hash generation, a FC layer is used to convert the video-level features into the decimal hash code, and the number of nodes in the FC layer is the length of hash code. Finally, the decimal hash code is transformed into the binary by the Sign function.

3.2 Generation of Pseudo Label

To enhance the performance of robustness and discrimination, we exploit the correlation distance of video-level features to generate pseudo-labels and guide the hashing learning.

After obtaining the features of each video in a batch, the correlation distances of these features are calculated pairwisely with the following equations:

$$\rho(\mathbf{Z}_i, \mathbf{Z}_j) = \frac{Cov(\mathbf{Z}_i, \mathbf{Z}_j)}{\sqrt{Var(\mathbf{Z}_i)}\sqrt{Var(\mathbf{Z}_j)}} \tag{4}$$

$$d(\mathbf{Z}_i, \mathbf{Z}_j) = 1 - \left|\rho(\mathbf{Z}_i, \mathbf{Z}_j)\right| \tag{5}$$

where $\mathbf{Z}_i$ and $\mathbf{Z}_j$ $(i, j = 1, 2, ..., n)$ are the two video-level features of two videos v_i and v_j, respectively, and n denotes the quantity of videos in the batch. $\rho \in [-1, 1]$ is the correlation coefficient, and the larger the absolute value of ρ, the higher perceptual similarity of the two videos. $Cov(\cdot)$ and $Var(\cdot)$ are the functions of covariance and variance, respectively. $d \in [0, 1]$ is the correlation distance.

To accurately describe the relationship of perceptual similarity, a histogram of the correlation distances d can be obtained by conducting mathematical statistics on n video-level features, as shown in Fig. 4. Based on the histogram, we calculate the mean μ_l and variance σ_l of the left half (i.e., $d \in [0, 0.5)$) of the histogram. Similarly, the mean μ_r and variance σ_r of the right part (i.e., $d \in [0.5, 1]$) can also be obtained. By analyzing the histogram distribution, the thresholds $t_l = (\mu_l - \sigma_l)$ and $t_r = (\mu_r - \sigma_r)$ are set to obtain the

pseudo-label matrix $\mathbf{S} \in \mathbb{R}^{n \times n}$:

$$\mathbf{S} = \begin{cases} S_{ij} = 1, & d(\mathbf{Z}_i, \mathbf{Z}_j) \leqslant t_l \\ S_{ij} = 0, & t_l < d(\mathbf{Z}_i, \mathbf{Z}_j) < t_r \\ S_{ij} = -1, & d(\mathbf{Z}_i, \mathbf{Z}_j) \geqslant t_r \end{cases} \tag{6}$$

If the distance is not greater than t_l, the label is assigned as 1, which means the two videos are predicted as perceptual identical. If the distance is not smaller than t_r, the label is assigned as -1, which means the two videos are predicted as perceptual distinct. If the distance is between t_l and t_r, it means it is not certain whether the perceptual contents are the identical or not, and the label is assigned as 0. Due to the scarcity of perceptual identical videos in the training set, there is an extremely rare possibility that $t_r \leqslant t_l$ or $t_l \leqslant 0$, in which case this batch does not participate in the training.

3.3 Objective Function

To further enhance the robustness and discrimination of the proposed video hashing, the model is optimized with an objective function $\mathcal{L}$ in a dual loss form, see Eq. (7).

$$\min \mathcal{L} = \mathcal{L}_1 + \alpha \mathcal{L}_2 \tag{7}$$

where $\mathcal{L}_1$ denotes the metric loss function, $\mathcal{L}_2$ denotes the classification loss function and α is a hyper-parameter.

We exploit the obtained pseudo-label matrix $\mathbf{S}$ as the ground truth for similarity judgment. Since the final binary hash code $\mathbf{c}$ is discrete, it makes objective function difficult to optimize the model with the standard backpropagation, thereby, we need to perform a relaxation. The loss function $\mathcal{L}_1$ is as follows:

$$\mathcal{L}_1 = \frac{1}{n^2} \sum_{i=1}^{n} \sum_{j=1}^{n} \left|S_{ij}\right| \left(\Phi_{ij} - S_{ij}\right)^2 + \eta \sum_{i=1}^{n} \|\mathbf{c}_i - \mathbf{h}_i\|_{\mathrm{F}}^2 \tag{8}$$

where $\{\mathbf{h}_i\}_{i=1}^n$, $\mathbf{h}_i \in \mathbf{R}^l$ and $\{\mathbf{c}_i\}_{i=1}^n$, $\mathbf{c}_i \in \{-1, 1\}^l$ denote the hash codes before and after the binarization, respectively, $\Phi_{ij} = \frac{1}{2} \left\langle \mathbf{h}_i, \mathbf{h}_j \right\rangle = \frac{1}{2} \mathbf{h}_i^{\mathrm{T}} \mathbf{h}_j$, $\|\cdot\|_{\mathrm{F}}^2$ denotes frobenius norm, and η is a hyper-parameter. The first term in Eq. (8) can keep the distance of identical features smaller than that of distinct features, and the second term can constrain the model to reduce errors in the binarization process.

In addition to the metric loss function, it is necessary to minimize the distance between identical videos and maximize the distance between distinct videos. For this purpose, we design a classification loss function $\mathcal{L}_2$ that can control the distance between identical videos and distinct videos in different magnitudes, see Eq. (9).

$$\mathcal{L}_2 = \frac{1}{n^2} \sum_{s_{ij} \in \mathrm{S}} - \left|S_{ij}\right| \left[\left(S_{ij} + 1\right) \cdot \log\left(p_{ij}\right) + \left(1 - S_{ij}\right) \log\left(1 - p_{ij}\right) \right] \tag{9}$$

where p_{ij} denotes the probability that the two videos v_i and v_j are predicted as perceptual identical.

The objective function with a dual loss $\mathcal{L}$ is used to conduct backpropagation of the model, and then generated continuous hash code are binarized to obtain the hash code $\mathbf{c}$. The procedure of our perceptual video hashing scheme is summarized in Algorithm 1.

Algorithm 1 The Proposed Video Hashing Scheme

Input: Training sample v, trainable model parameters θ, the deep video hashing network f, testing sample $\mathcal{X}$, batchsize n

while $t \leq epoches$ **do**
  for $i = 1 : n$ **do**
    for $j = 1 : n$ **do**
      $d_{ij} \leftarrow d\left(f_\theta\left(v_i\right), f_\theta\left(v_j\right)\right)$
  $t_r, t_l \leftarrow$ adaptive perceptual similarity threshold
  Generate pseudo-label matrix $\mathbf{S}$ by Eq. (6) and t_r, t_l
  $\mathcal{L}_1 = \frac{1}{n^2} \sum_{i=1}^{n} \sum_{j=1}^{n} \left|S_{ij}\right| \left(\Phi_{ij} - S_{ij}\right)^2 + \eta \sum_{i=1}^{n} \|\mathbf{c}_i - \mathbf{h}_i\|_{\mathrm{F}}^2$
  $\mathcal{L}_2 = \frac{1}{n^2} \sum_{s_{ij} \in \mathrm{S}} - \left|S_{ij}\right| \left[\left(S_{ij} + 1\right) \cdot \log\left(p_{ij}\right) + \left(1 - S_{ij}\right) \log\left(1 - p_{ij}\right) \right]$
  $\mathcal{L} = \mathcal{L}_1 + \alpha \mathcal{L}_2$
  Update the parameters θ to minimize $\mathcal{L}$
  $t = t + 1$

Out-of-Sample-Extension: Feed the testing sample $\mathcal{X}$ into the network f and calculate the output $f_\theta\left(\mathcal{X}\right)$

Hash code: Binarize $f_\theta\left(\mathcal{X}\right)$ to $\mathbf{c}$

4 EXPERIMENTAL RESULTS AND ANALYSIS

4.1 Experimental Setup

The first two columns of Table 1 list seven kinds of content-preserving manipulations with corresponding parameters, by which each original video can produce 20 perceptual identical videos (i.e., copied version). As for the manipulation of frame substitution means that, a frame different from original video content was added at the time step of the dropped frame, and the number ratio of inserted irrelevant frames to the original frames in the video was 10%. To demonstrate the generalization ability, we randomly selected videos from Kinectics [21] and TRECVID [22] as training data and testing data. Table 2 shows the structure of training set and testing set.

Table 1: Statistics of Hash Distances under Different Content-preserving Video Manipulations

Manipulation	Parameter	Statistic			
		Max.	Min.	Mean	Std.
Salt & pepper noise	0.01, 0.03, 0.05, 0.1	5.65	0	2.01	0.38
Gaussian noise	0.01, 0.03, 0.05	4.90	0	1.98	0.27
Logo blocking	80×80	3.46	0	1.14	0.19
Frame rotation	15°, 30°, 45°	5.29	0	1.99	0.31
Median filtering	3×3, 5×5, 7×7, 10×10	2.82	0	0.73	0.18
Frame scaling	0.8, 0.9, 1.1, 1.2	4.00	0	1.20	0.14
Frame substitution	10%	5.29	0	2.29	0.35

Table 2: The Structure of Datasets

	Training dataset	Testing dataset
Dataset	Kinectics	TRECVID
Original videos	1,500	400
Identical videos	1500×20=30,000	400×20=8,000
Total	31,500	8,400

The hyperparameters α and η were set to 0.5 and 2, respectively, and the discussion is given in Section 4.4. The length of the hash code defaults to 64 bits. For the gradient descent, we used Adam optimizer with a learning rate of 0.001, the mini-batch of 128 and the weight decay of 0.0002. In the following experiments, Euclidean distance was utilized to calculate the distance between two hashes.

4.2 Perceptual Robustness and Discrimination

Perceptual robustness means that, the hash code of an original video should be similar to those of perceptual identical videos as

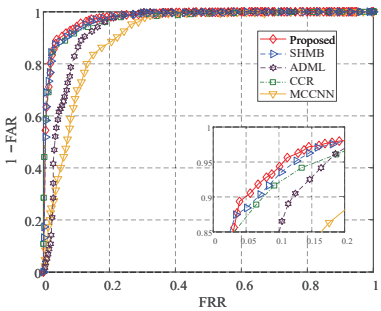
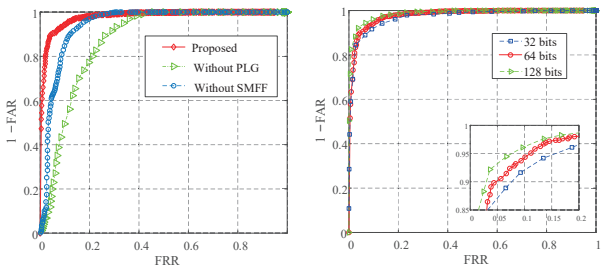


Figure 5: Performance comparisons of copy detection for our scheme, [23], [24], [25] and [15].



(a) Different model strategies (b) Different hash lengths

Figure 6: ROC curves under different model strategies and hash lengths.

much as possible. After conducting the content-preserving manipulations, the hash distances between each original video and its corresponding perceptual identical videos were calculated, see the 4th to 6th columns of Table 1. All hash distances are kept within a small range, which means that the proposed scheme can effectively resist these attacks with perceptual robustness. It is noted that all hash distances do not exceed 5.65, thereby, this value can be used as a candidate value of the threshold T in Eq. (1).

The discrimination capability of the proposed scheme was also evaluated. We calculated the hash distances of each original video between the other perceptual distinct videos in the testing set. Thus, we can obtain a total of $(400 \times 399)/2 = 79,800$ distances for all distinct video pairs. From the statistical results of hash distances between perceptual distinct videos, we can find that the mean value is 11.29 and the standard deviation is 5.37. Compared with the statistics of robustness results in Table 1, we can see that the hash distance of distinct videos is significantly larger than that of identical videos, thus proving the discrimination performance of our scheme. Overall, our scheme has satisfactory robustness and discrimination, and can be effectively used for video copy detection.

4.3 Performance Comparison

We selected 300 original videos with different perceptual contents to compare with other schemes. Also, for each original video, 20 perceptual identical videos were generated according to the manipulations in Table 1. Thus, we can obtain $C_{300}^2 = 44,850$ hash distances of perceptual distinct video pairs. By the same token, we can also obtain $C_{21}^2 \times 300 = 63,000$ hash distances of perceptual identical video pairs.

For a perceptual robust video hashing scheme, the determination of the threshold T in Eq. (1) can greatly affect judgment accuracy for perceptual identical and distinct videos. By setting different thresholds T, we can calculate a series of false acceptance rates (FAR) and false rejection rates (FRR) via Eqs. (10) and (11).

$$\mathrm{FAR}(T) = P(D(\mathbf{c}_1, \mathbf{c}_2) < T \mid v_1 \text{ and } v_2 \text{ are distinct pair}) \quad (10)$$

$$\mathrm{FRR}(T) = P(D(\mathbf{c}_1, \mathbf{c}_2) \geq T \mid v_1 \text{ and } v_2 \text{ are identical pair}) \quad (11)$$

where $\mathbf{c}_1$ and $\mathbf{c}_2$ are the hash codes of the two videos v_1 and v_2, respectively. Obviously, the lower values of FAR and FRR, the better the judgment accuracy is achieved. Thus, the receiver operating characteristic (ROC) curve with FRR and (1−FAR) can be drawn to evaluate the overall performance of copy detection with respect to perceptual robustness and discrimination.

In the experiment, four typical schemes were chosen for performance comparison, including SHMB [23], ADML [24], CCR [25] and MCCNN [15]. For MCCNN [15], we used the average result of hash codes for all frame images to represent the video hash code. To guarantee the fairness of the comparisons, we tuned all these schemes to their optimal performance. Note that, the lengths of hash code for [23], [24], [25] and [15] were defaultly set as 96 bits, 100 bits, 256 bits and 75 bits, respectively, and our scheme was 64 bits. As shown in Fig. 5, the ROC curve of our scheme is the closest to the upper left corner, which means our scheme achieves the best overall performance with respect to perceptual robustness and discrimination. Although the effect of SHMB [23] is approximate to ours, it is worth noting that SHMB scheme is supervised, and its hash length is 96 bits. That is to say, our video hashing scheme not only has excellent performance for copy detection, but also greatly saves labor cost, while the hash code has low redundancy, saving storage and computation cost.

Besides, F_1 score was also utilized as an evaluation criteria for the accuracy of copy detection, and a larger F_1 score means better performance, see Eq. (12).

$$F_1 = \max_T \left\{ \frac{2 \cdot [1 - \mathrm{FAR}(T)] \cdot [1 - \mathrm{FRR}(T)]}{[1 - \mathrm{FAR}(T)] + [1 - \mathrm{FRR}(T)]} \right\} \quad (12)$$

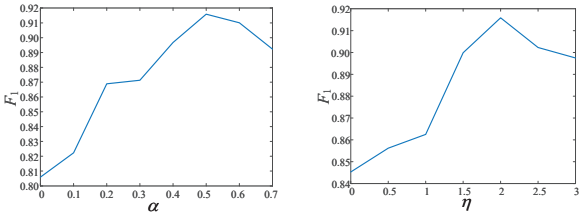
To illustrate the superiority of our unsupervised scheme, we added two untrained content-preserving manipulations, i.e., gamma correction and average filtering with the parameters 0.75, 0.95, 1.25 and 3×3, 4×4, 5×5, respectively. Table 3 lists comparison results of F_1 score on different video manipulations. We can find that, for most manipulations, the F_1 scores of our scheme are greater than those of other compared schemes. It should be noted that the performance of our scheme does not degrade much in the case of untrained manipulations, which indicates the strong generalization ability.

4.4 Ablation Study

Our scheme contains two main strategies, i.e., the 3D self-attention, multi-scale feature fusion model (denoted as SMFF) and the pseudo-label generation (denoted as PLG). In order to verify the effectiveness of these two strategies, we replaced the former with R(2+1)D [26], and maintained the other parts. As for the PLG, we discarded this strategy and directly used the hash codes produced by the SMFF. We trained these two ablation strategies and compared them with the default proposed model. From Fig. 6(a), it can be found that the two strategies both lead to a certain gain.

Table 3: Comparisons of F_1 Score on Video Manipulations

Manipulation	Proposed	SHMB	ADML	CCR	MCCNN
Salt & pepper noise	0.9129	0.9076	0.8813	0.8843	0.9041
Gaussian noise	0.9281	0.9203	0.8617	0.9186	0.8937
Logo blocking	0.9003	0.8832	0.8379	0.9014	0.8764
Rotation	0.9589	0.9570	0.8934	0.9478	0.8971
Median filtering	0.9999	0.9714	0.9345	0.9989	0.9389
Frame scaling	0.9787	0.9856	0.8910	0.9834	0.9654
Frame substitution	0.9216	0.9236	0.8211	0.9104	0.6798
Gamma correction	0.9244	0.8711	0.8089	0.8507	0.7156
Average filtering	0.9178	0.8334	0.7901	0.8176	0.7478
Average	0.9159	0.8813	0.8492	0.8754	0.8313



(a) Parameter α (b) Parameter η

Figure 7: Performance of F_1 score under different values of hyper-parameters α and η.

We also conducted the performance analysis of our scheme under different hash lengths. In order to realize hash generation with different lengths, we modified the number of nodes in the last FC layer without changing other conditions. It can be seen from Fig. 6(b), the ROC curve of 32 bits is the worst, and the ROC curve of 128 bits is the best, but it is not much different from that of 64 bits. In view of the complexity of the computation, the hash length of 64 bits is recommended.

In addition, we analyzed the sensitivity of hyper-parameters α in Eq. (7) and η in Eq. (8) on the performance of our video hashing. We first set α as 0 to test the sensitivity of η, and vice versa. Fig. 7 shows the influences of hyper-parameters α and η on the F_1 score of our scheme. We can find that, when α is set as 0.5, the $\mathcal{L}_2$ loss can bring a obvious improvement that is better than other parameter values; when η is set as 2, the performance of our video hashing reaches the best at this situation.

5 CONCLUSION

In this work, we propose an unsupervised video hashing scheme that exploits 3D-CNN to extract the rich spatio-temporal information of videos. The Inception block and the 3D self-attention mechanism are designed to integrate different levels of information, thus increasing the ability of extracting features. In the training stage, to construct the pseudo-labels based on the similarity relationship of perceptual contents, we perform statistical analysis and calculate the correlation distance on the video-level features. Since our scheme does not require additional supervised information, it not only has superior copy detection performance compared with some state-of-the-art schemes, but also is more suitable for the increasingly massive big data.

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